

Wireless Implant Communications for Biomedical Monitoring Sensor Network

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Abstract—Galvanic coupling provides a novel data transmission between sensor units for low frequency intra-body communication with electrodes attached to the human skin. In this work, that approach has been adapted to implantable miniaturized pills. A communication system for wireless data transmission in muscle tissue is developed capable of transmitting data on four channels concurrently with a throughput of 4.8 kbit/s. The main focus is the future implantability of such a miniaturized system for medical long term surveillance of patients. To achieve this goal, circuit size, low power consumption and electrical safety have to be carefully considered. The implemented frequency division multiple access (FDMA) system works in the frequency range between 100 kHz and 250 kHz. Tests have been processed with 4 transmitters units. The system architecture features the possibility of migration into a single system-on-chip.

I. INTRODUCTION

The latest monitoring devices for biomedical application offer the possibility to observe a patient's long term conditions. Such monitoring units are worn or even implanted directly into human tissue [1], [2]. The communication between devices is established by wireless transmission over on-body or intra-body networks. Recent studies focus on intra-body communication where implanted miniaturized pills are transmitter and receiver [3], [4]. For communication between two of those pills, galvanic coupling has been investigated in [5], [6]. These approaches couple alternating current within the range of 10 kHz to 1 MHz by a pair of differential electrodes into the human tissue. For low frequency communication with electrodes attached to the human skin, galvanic coupling provides a reliable connection between transmitter and receiver.

The presented system uses the same technology but instead of skin-mounted electrodes, the differential electrodes build the two ends of a pill-shaped implant (Fig. 1). Two versions have been tested, one with a cylindrical copper ending (length 1 cm and diameter 4 mm), the other one with just the diameter of the pill as contacting surface. Since muscle tissue is quite conductive compared to skin tissue, it has been assumed that the losses between the differential electrode pairs are decreased comparing on-body electrodes and pill-shaped implants.

The goal of the communication system is a reliable, low-power data link between up to 4 transmitters with one receiver unit. A data rate of 4.8 kbit/s allows a sensor to transmit real time data for monitoring human vital parameters, e.g. temperature, SpO_2 , ECG. The transmission frequency has been chosen between 100 kHz and 250 kHz due to previous feasibility measurements [7].

II. ARCHITECTURE

For simplicity reason, the first system approach considers an unidirectional communication system with 4 transmitters in star topology and a central receiver. A frequency division multiple access (FDMA) system has been realized. This offers more freedom for transmitter separation compared to time division multiple access (TDMA). The latter has not been chosen due to the fact that slot negotiation and synchronization would require more efforts in synchronization and a bidirectional full-duplex communication. The carrier frequencies of the 4 channels in this application are located at 100, 150, 200 and 250 kHz.

The chosen differential binary phase shift keying (DBPSK) transmission scheme is a very simple modulation scheme. It consists of a differential encoding and consecutive a binary phase-shift modulation.

A. Differential Encoding

The data stream is encoded differentially. Every binary '0' data bit flips the state of the last encoded bit, every '1' leaves it unchanged and the same value is encoded again. This way the information is encoded in the phase shift of the carrier wave rather than the phase itself. The process is illustrated in Tab. I.

The transmission is independent of the signal polarity. Hence the orientation of the implants does not affect the transmission. But much more freedom is offered to the surgeon for placing the units in the body.

In addition, an overhead is induced by transmitting a hamming code for detection and correction one faulty received bit. Assuming the transmission of 8 bit data words, four

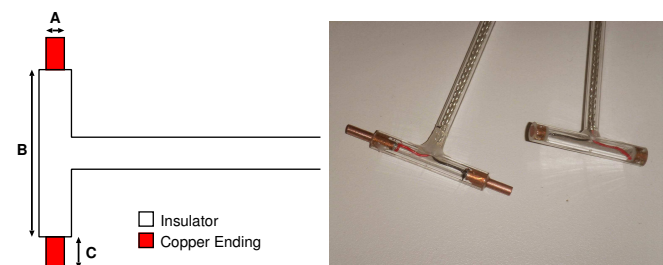


Fig. 1. Image of the used electrodes on the right hand side, including a schematic drawing on the left hand side. The two used versions have the dimensions: $A = 0.4$ cm, $B = 5.0$ cm, $C = 1.0$ cm for the longer electrode contacts and $C = 0.0$ cm for the short electrode contacts respectively

TABLE I
GENERATION OF DIFFERENTIALLY ENCODED DATA: THE ENCODED
SEQUENCE IS INITIALIZED BY 1

Data sequence	Init	1	0	0	1	0	0	1	1
Encoded sequence	1	1	0	1	1	0	1	1	1
Transmitted phase	0	0	π	0	0	π	0	0	0

additional hamming bits have to be included into the data stream. Therefore, a transfer speed of 7.2 kbit/s is required to achieve the desired data throughput of 4.8 kbit/s.

B. Modulation Scheme

Binary phase shift keying (BPSK) was chosen due to be more immune to amplitude variations and external noise. Furthermore it offers relatively low complexity for modulation and demodulation. The encoded information sequence is modulated with the carrier frequency of the corresponding transmitter unit. In DBPSK, the transmission of an encoded '1' results in no phase shift, while a '0' shifts the phase by 180 degrees.

C. Pulse shaping

The signal to noise ratio of each channel has been improved by shaping the modulated signal with a window function. The window length is given by the length of a transmitted bit.

Ideally the *sinc* function

$$A_{sincpulse} = sinc(x) = \frac{\sin(\pi \cdot x)}{\pi \cdot x} \quad \text{for } -1 \leq x \leq 1 \quad (1)$$

transforms to a rectangle in Fourier space. Due to hardware limitations, the filter length for the pulse shaping is limited. The pulse length matches the number of transmission samples per bit. With a sample rate of 1 MHz and a bit rate of 7.2 kbit/s one bit pulse contains 140 samples. Samples of the originating and the pulse shaped waveforms are plotted in Fig. 3.

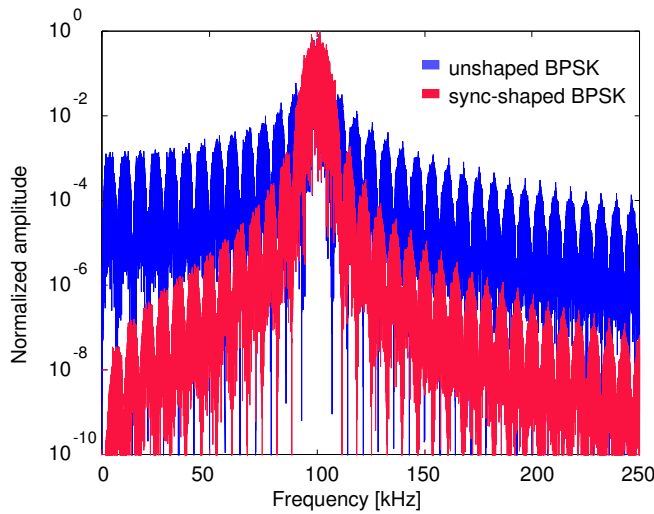


Fig. 2. Normalized spectrum of BPSK modulated wave: With and without pulse shaping using a center frequency of 100 kHz, a bit rate 7.2 kbit/s and a sample rate 1 MHz

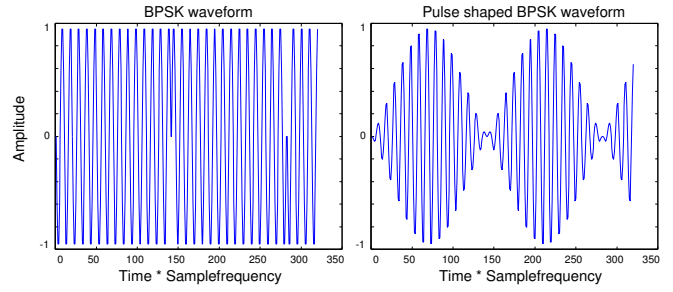


Fig. 3. Waveform of BPSK modulated signals. On the left hand, no pulse shaping is applied. On the right hand, the bit pulses are *sinc* shaped. Center frequency is 100 kHz, bit rate 7.2 kbit/s, sample rate 1 MHz

The *sinc* function, which forms the pulse shape, is cut at -1 and 1 according to Eq. 1. Therefore the spectrum of *sinc* shaped BPSK shown in Fig. 2 is not ideally rectangular shaped, but shows the typical side lobes. Regarding a signal bandwidth of 50 kHz per channel the spectrum of the pulse shaped signal at $f_{center} \pm 25$ kHz is more than 20 dB lower than the unshaped BPSK spectrum. Therefore, the pulse shaping results in a significant improvement of the signal to noise ratio if multiple transmitters are active simultaneously.

D. Demodulation

On the receiver side, the incoming signal is mixed down to baseband and low-pass filtered to retrieve the original pulses. The phase information is extracted from the inphase (real) and quadrature (imaginary) part to obtain the phase information needed for decoding. The receiver synchronizes on the bit pulses and differentially decodes incoming data by calculating the phase of each pulse.

III. SYSTEM DESIGN

Since this communication system is planned to be used in a biomedical application, it has to be safe for the patient. No interference with the patient's body functions is allowed, e.g. the nervous system. Therefore, it is not allowed to emit signals with a frequency below 10 kHz since body signals are located in that frequency range.

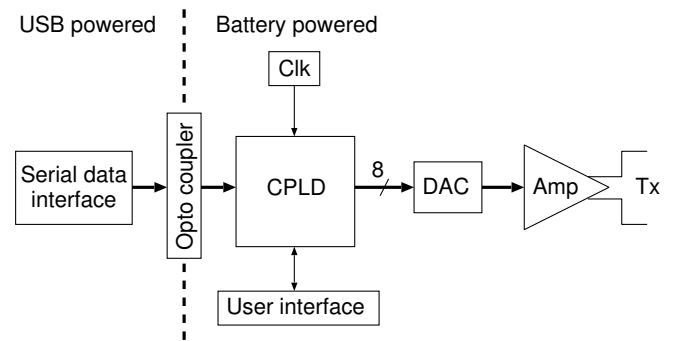


Fig. 4. Schematic view of the design blocks of the transmitter: Serial data is DBPSK modulated and differentially coupled by a current amplifier

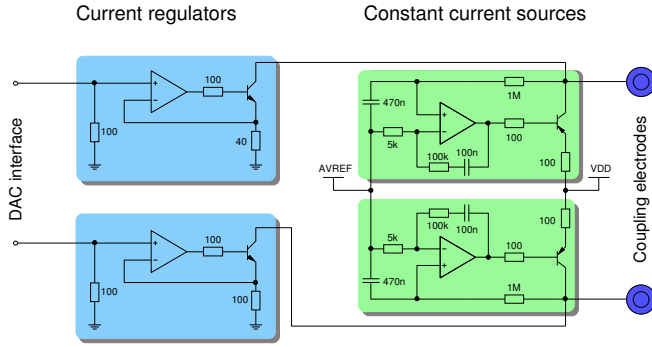


Fig. 5. Architecture of the differential current amplifier: Current regulator driven by the DAC (left part) and constant current sources (right part)

Special care has been taken of low power design since it is planned to be implemented on a mixed signal chip in the future. The system should be able to run over a longer time period without recharging.

Electrical decoupling of the sensor from any further power nets is realized with opto couplers and battery powering. The transmitter and the receiver units are composed of 3.3 V devices. The size of the transmitter board is 16 cm² while the receiver occupies 29.5 cm².

A. Functional Blocks of the Transmitter

The transmitter is composed of very few operational units (Fig.4). Despite the fact that CPLDs only provide a very small number of functional blocks and logic cells it has been the choice to converge the implementation toward a future ASIC integration.

The operating voltage is 3.3 V. The highest carrier frequency of 250 kHz defines the system clock of 1 MHz enabling a four times oversampling. The digital part sends data by an 8 bit digital-to-analog converter (DAC) to the analog output driver. A parallel data interface for the DAC is used because a serial input would require an eight times higher system clock frequency for data transmission resulting in a higher power dissipation. In order to save energy while no data is sent, the entire analog part can be shut down.

The transmitter has two symmetrical branches (Fig. 5). Each consists of a current regulator controlled by the DAC and a constant current source. It generates a differential current of 1 mA between the two electrode contacts, which means that one of the two electrode contacts is supplying while the other one is sinking.

Additionally, a serial interface from a computer configures the digital unit and provides data for transmission.

B. Functional Blocks of the Receiver

Figure 6 shows the main functional blocks of the receiver with analog input amplifier and filter, ADC, FPGA for digital demodulation and interfaces.

The input amplifier of the receiver has to amplify very weak signals to a level resolvable by the following ADC. The amplitude of the input signal goes from about 5 mV at 5 cm

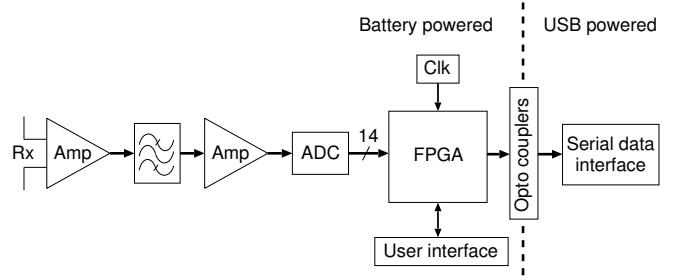


Fig. 6. Schematic view of the design blocks on the receiver: Received signals are amplified, bandpass filtered, digitally demodulated and encoded for serial data transmission

distance between transmitter and receiver electrodes, down to about 5 μV at 45 cm distance. Primary goal in the design of the amplifier is a minimal induced noise; secondary goal is a low power consumption.

The amplifier is built in three stages (Fig. 7). Since only the AC part of the signal is needed for data communication, the first stage has to provide a maximal common mode rejection ratio (CMMR) to reject the incoming DC level. The most suitable active amplifier structure is an instrumentation amplifier because of its differential signal branches. Each of the amplifier sections is designed to amplify the incoming signal by a factor of 10. In addition, the second stage of the amplifier contains an active bandpass filter. The third stage contains a non inverting amplifier, its purpose is the final amplification of the input signal to optimally make use of the ADC resolution. It has an amplification factor of 10, which results in a total amplification of 60 dB.

The bandpass filter has been implemented as a Sallen-Key structure (Fig 7, middle part), The values of the filter elements can be calculated according to the following formulas:

$$C1 = 2 \cdot C2 \quad (2)$$

$$R = \frac{\sqrt{2}}{4 \cdot \pi \cdot f_c \cdot C2} \quad (3)$$

where f_c is the 3 dB cutoff frequency and the values of $C1$ and $C2$ are chosen to match the ratio 1:2 as good as possible. The cutoff frequencies are 75 kHz for the high pass and 350 kHz

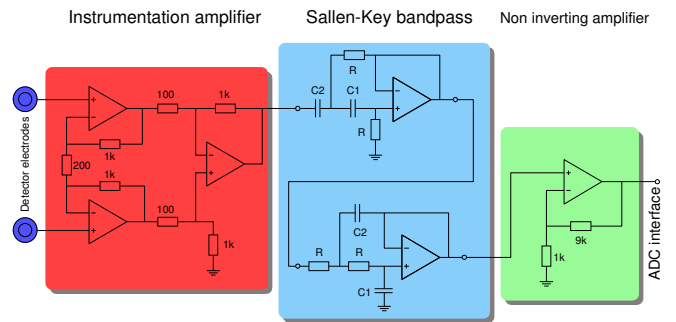


Fig. 7. Input amplifier stages: Instrumentation amplifier structure (left part), second order Sallen-Key filters (middle part top: high pass, bottom: low pass) and final non-inverting amplifier (right part)

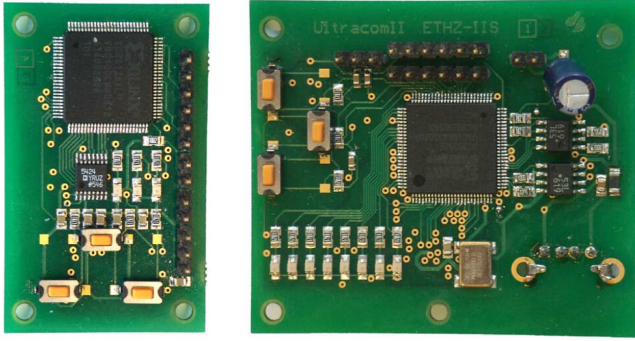


Fig. 8. Transmitter and receiver PCB: 4 layers, 16 cm² respectively 29.5 cm²

for the low pass. An analog-to-digital converter (ADC) with 14 bits resolution converts the signals.

The digital part of the receiver (Fig. 6) processes the demodulation algorithm and allows different receiver modes for data detection and recording.

Finally, a serial data interface transmits the received data to a computer or any receiver circuit for storage or examination. In addition, the interface enables the configuration of the FPGA. The built-in digital clock managers, multipliers and data storage structures of the FPGA offer convenient features for the needed data encoding algorithm. The FPGA based prototype shown in Fig. 8 allows migration of the architecture into a system-on-chip (SoC).

IV. DISCUSSION

It has not been feasible to implant the test devices into a human body to conduct the measurements for this study. Instead an elliptical phantom with a size of $60 \times 40 \times 10.5$ cm³ has been used. It was filled with muscle-tissue simulating liquids *MSL27*, a liquid which simulates the electromagnetic properties of muscle tissue at 27 MHz. There has not been any liquid available designed for the employed lower frequency range. The transmitter and the receiver electrodes were inserted 5 cm into the liquid and attached to non conducting holders. The tests have been performed inside an anechoic chamber reducing environmental noise to a minimum level.

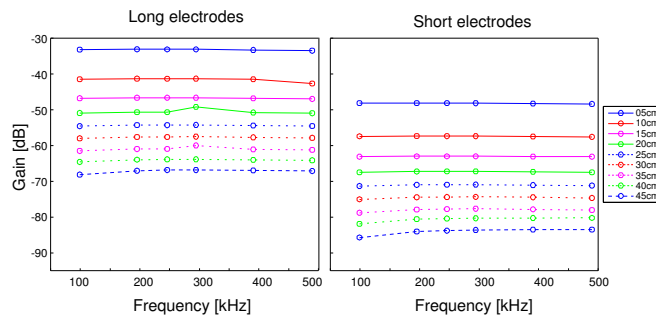


Fig. 9. Measured attenuation of the galvanic coupled electrodes with long and short contacts at different distances

The measurement results for parallel aligned transmitter and receiver electrodes look promising. The main difference between the long poles and the short ones is a 15 dB weaker input signal (Fig. 9).

The current output of the transmitter unit drives 1 mA up to a load of 1 k Ω . This is by far sufficient considering coupling measurement of muscle tissue. It is feasible to integrate the transmitter unit as ASIC for miniaturization to implantable size of the pill electrodes (Fig. 1). Even with the preliminary approach of discrete elements, the total transmitter unit has a power consumption of 16.5 mW only.

The receiver is able to demodulate the data of a single transmitter up to a distance of 45 cm between the transmitter and receiver electrodes. In case of multiple transmitters, the performance of the receiver could be improved with an advanced analog part offering band selective and adaptive gain so that every FDMA channels can be equalized independently. However, that will increase the complexity and the power consumption of the receiver unit.

V. CONCLUSION

Based on galvanic coupling, a data communication system is presented demonstrating the feasibility for wireless communication of implants in muscle tissue. The sensor network consists of 4 transmitters and one receiver. The signals of the transmitters are in the frequency range between 100 kHz and 250 kHz and have a peak current of 1 mA. The data is differentially encoded and BPSK modulated.

The low system clock frequency and the small logic complexity allow for an efficient low-power design suitable for implant communication in biomedical monitoring applications, e.g. temperature, SpO₂ and ECG sensors. Transmitter and receiver are built by several discrete components which are dimensioned to allow straight forward integration into a mixed-signal SoC.

REFERENCES

- [1] T. Handa, S. Shoji, S. Ike, S. Takeda, and T. Sekiguchi, "A very low-power consumption wireless ECG monitoring system using body as a signal transmission medium," in *Proceedings of the International Conference on Transducers, Solid-State Sensors and Actuators*, 1997, pp. 1003–1007.
- [2] D. P. Lindsey, E. L. McKee, M. L. Hull, and S. M. Howell, "A new technique for transmission of signals from implantable transducers," *IEEE Transactions on Biomedical Engineering*, vol. 45, no. 5, pp. 614–619, 1998.
- [3] H. Sawan, Y. Hu, and J. Coulombe, "Wireless smart implants dedicated to multichannel monitoring and microstimulation," *IEEE Circuits and systems magazine*, pp. 21–39, 2005.
- [4] J. H. Schulman, J. Mobley, J. Wolfe, H. Stover, and A. Krag, "A 1000+ channel bionic communication system," in *28th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC06)*, New York, USA, September 2006, pp. 4333–4335.
- [5] N. Yao, H.-N. Lee, R. Scabassi, and M. Sun, "Low power digital communication in implantable devices using volume conduction of biological tissues," in *IEEE EMBC'06, New York, USA*, 2006, pp. 6249–6252.
- [6] M. S. Wegmueller, M. Oberle, N. Felber, N. Kuster, and W. Fichtner, "Galvanical coupling for data transmission through the human body," in *IEEE IMTC, Sorrento, Italy*, 2006, pp. 1686–1689.
- [7] M. Hediger and T. Kaufmann, "Wireless implant communications," Diploma thesis, IIS/ETH Zurich, 2006.